Microsoft Elevate

capstone project

Project Title:

Consumer Financial Complaints Analysis

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Outline

Problem Statement (Should not include solution)

Proposed System/Solution (Project Background)

Proposed System/Solution (Data Preparation)

System Development Approach (Technology Used)

Result – Complaint Overview Dashboard (Page 1)

Result – Complaint Issue Analysis Dashboard (Page 2)

Result – Dispute & Resolution Analysis Dashboard (Page 3)

Key Insights, Recommendations & Future Scope

Conclusion

References

Problem Statement

Consumer financial institutions receive a **large number of complaints** from customers regarding issues such as **credit reporting errors, debt collection practices, loan servicing problems, and unauthorized transactions**. These complaints are collected and published by regulatory authorities like the **Consumer Financial Protection Bureau (CFPB)** to ensure transparency and consumer protection. However, due to the large volume and variety of complaints, it becomes difficult to identify patterns, understand the most common issues, and determine **which financial products generate the highest number of consumer grievances. Without systematic analysis**, important insights related to complaint trends, dispute behavior, and company responses may **remain hidden**.

Therefore, analyzing consumer complaint data to identify major issues, complaint patterns, and areas of consumer dissatisfaction becomes an important task for improving financial service quality and regulatory oversight.

Proposed System/Solution (Project Background)

The proposed proposed aims to address the **challenge of understanding patterns** in consumer financial complaints and **identifying major areas of consumer dissatisfaction** in the financial services sector. The Consumer Financial Protection Bureau (CFPB) publishes a large dataset containing complaints submitted by consumers regarding financial products such as **credit cards, loans, mortgages, and debt collection services**. However, due to the **large volume and complexity** of this data, it becomes difficult for stakeholders to quickly identify key issues, complaint trends, and companies receiving the highest number of complaints.

To address this challenge, the proposed solution focuses on **analyzing the CFPB complaint dataset** to extract meaningful insights that can help regulators and organizations better understand consumer concerns. The system aims **to organize and examine complaint data to highlight patterns** related to financial products, **types of issues reported by consumers**, and the **effectiveness of company responses** to these complaints. By systematically studying this data, it becomes possible to **identify recurring problems** and areas where **consumers frequently experience dissatisfaction**.

The proposed approach emphasizes transforming **raw complaint records** into **structured analytical information** that can support decision-making. Through this analysis, stakeholders can gain a clearer understanding of **complaint distribution across financial products**, **common consumer issues, and dispute behaviors**. Ultimately, the proposed system provides a structured method for examining consumer complaint data to reveal insights that may contribute to **improving financial service quality** and **consumer protection efforts**.

Proposed System/Solution (Data Preparation)

The dataset used in this project is the Consumer Financial Complaints dataset published by the Consumer Financial Protection Bureau (CFPB), which contains complaint records related to various financial products and services.

Data Cleaning:
During the preparation process, missing values were handled by marking unspecified fields as **“Not Specified”** where appropriate. Rows containing critical missing values in key analytical fields such as product or issue categories were reviewed and filtered when necessary.

Column Selection:
Columns that were not required for analytical purposes, such as narrative text and certain auxiliary fields, were removed to simplify the dataset and improve processing efficiency.

Feature Preparation:
Additional analytical attributes were created to support further analysis. For example, the **complaint year was extracted from the complaint date** to enable trend analysis over time.

Data Structuring:
The cleaned dataset was organized in a structured format, ensuring that relevant fields such as product category, issue type, company response, and dispute status were ready for analytical processing.

System Development Approach (Technology Used)

Data Source:
The CFPB Consumer Financial Complaints dataset was used as the primary data source for this project. The dataset contains records of consumer complaints related to financial products, issues reported by consumers, and company responses.

Data Preparation using SQL:
Before visualization, the dataset was processed using **PostgreSQL** to ensure that the data was clean and structured for analysis. During this stage, missing values were handled, unnecessary columns were removed, and additional analytical attributes such as **complaint year** were derived from the complaint date.

Power BI Data Modeling:
The prepared dataset was imported into **Power BI Desktop**, where the data model was created. A **denormalized flat complaints table** was used to simplify reporting and analysis. Additionally, a **Date Dimension table** was created and linked to the complaint date field to support time-based analysis.

Measure Creation and Analysis:
Several **DAX measures and calculated fields** were created to calculate key indicators such as total complaints, dispute rate, complaint trends, and complaint distribution by product and company.

Dashboard Development:
Interactive visualizations such as **bar charts, matrices, trend charts, and KPI cards** were developed to present insights in an intuitive and user-friendly manner.

Deployment:
The completed Power BI dashboard was **published to Power BI Service**, enabling the report to be shared and accessed through an interactive web-based reporting environment.

Result – Complaint Overview Dashboard (Page 1)

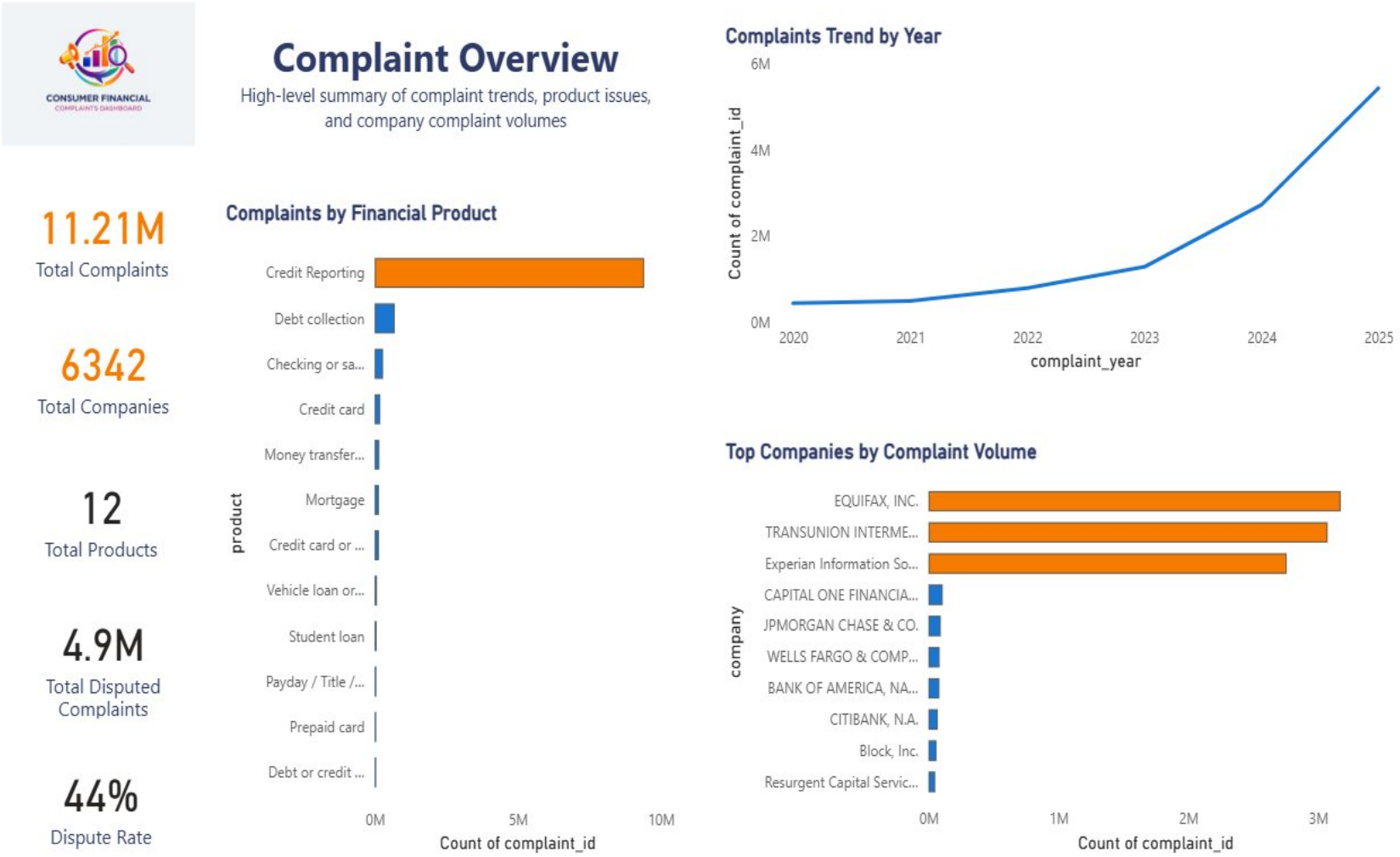
Key Dashboard Insights:

This dashboard provides a high-level overview of consumer financial complaints reported to the CFPB.

Key insights include:

- Total number of complaints received between 2020–2025
- Complaint trends over time to identify growth in consumer issues
- Financial products generating the highest complaint volumes
- Companies receiving the most consumer complaints

This overview helps stakeholders quickly identify **major complaint drivers and high-risk financial products**.



Result – Complaint Issue Analysis Dashboard (Page 2)

Key Dashboard Insights:

This dashboard analyzes the types of problems consumers report when filing complaints.

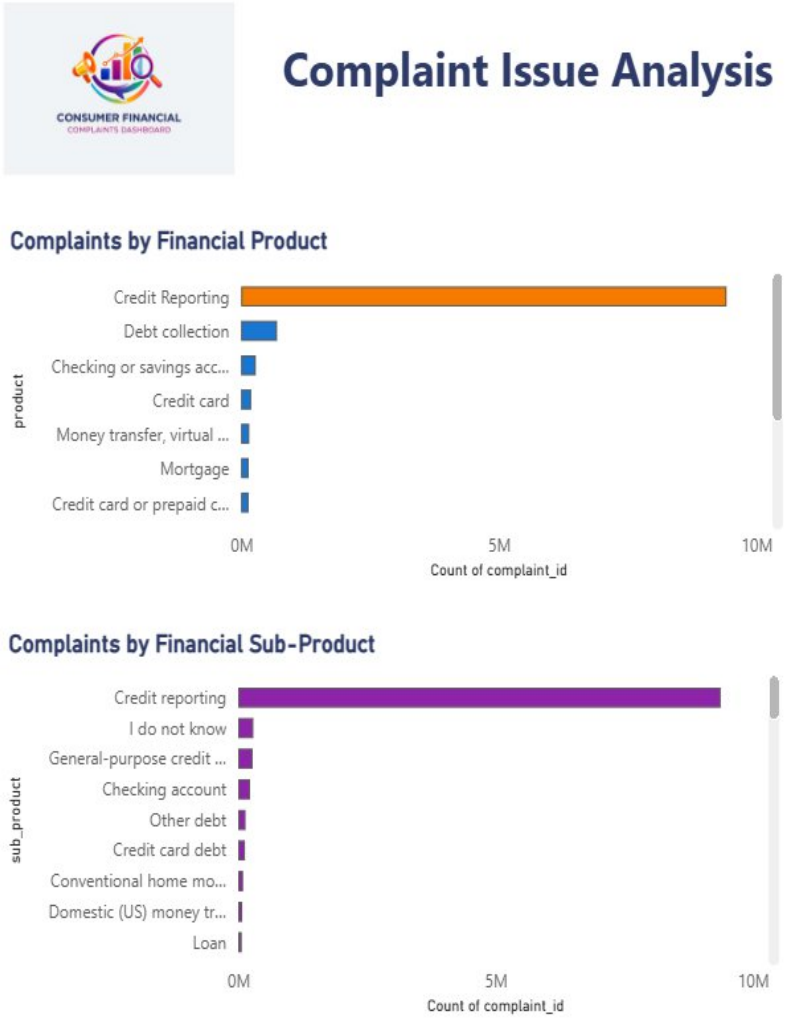
The analysis highlights:

- Most frequently reported consumer complaint issues

- Complaint distribution across financial products and sub-products

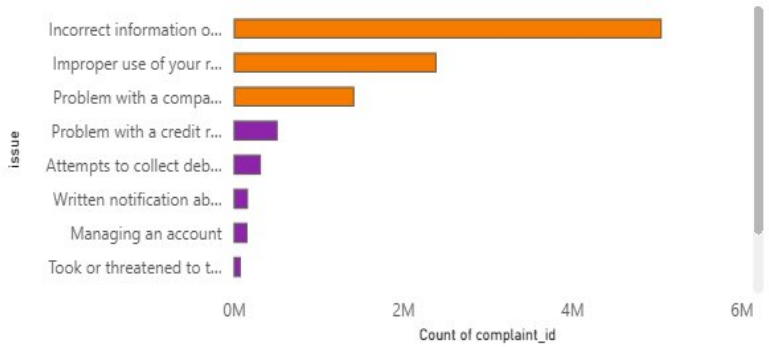
- Relationships between financial products and specific complaint issues

This helps identify **systemic problems in financial services such as inaccurate credit reporting and debt collection issues**.



Product vs Issue Breakdown

product	Attempts to collect debt not owed	False statements or representation	Improper use of your report
Checking or savings account			235
Credit card			1896
Credit card or prepaid card			1109
Credit Reporting			2389383
Debt collection	320143	75652	
Mortgage			565
Payday / Title / Personal Loan			750
Student loan			1842
Vehicle loan or lease			1240
Total	**320143**	**75652**	**2397020**



Result – Dispute & Resolution Analysis Dashboard (Page 3)

Key Dashboard Insights:

This dashboard evaluates how effectively companies resolve consumer complaints.

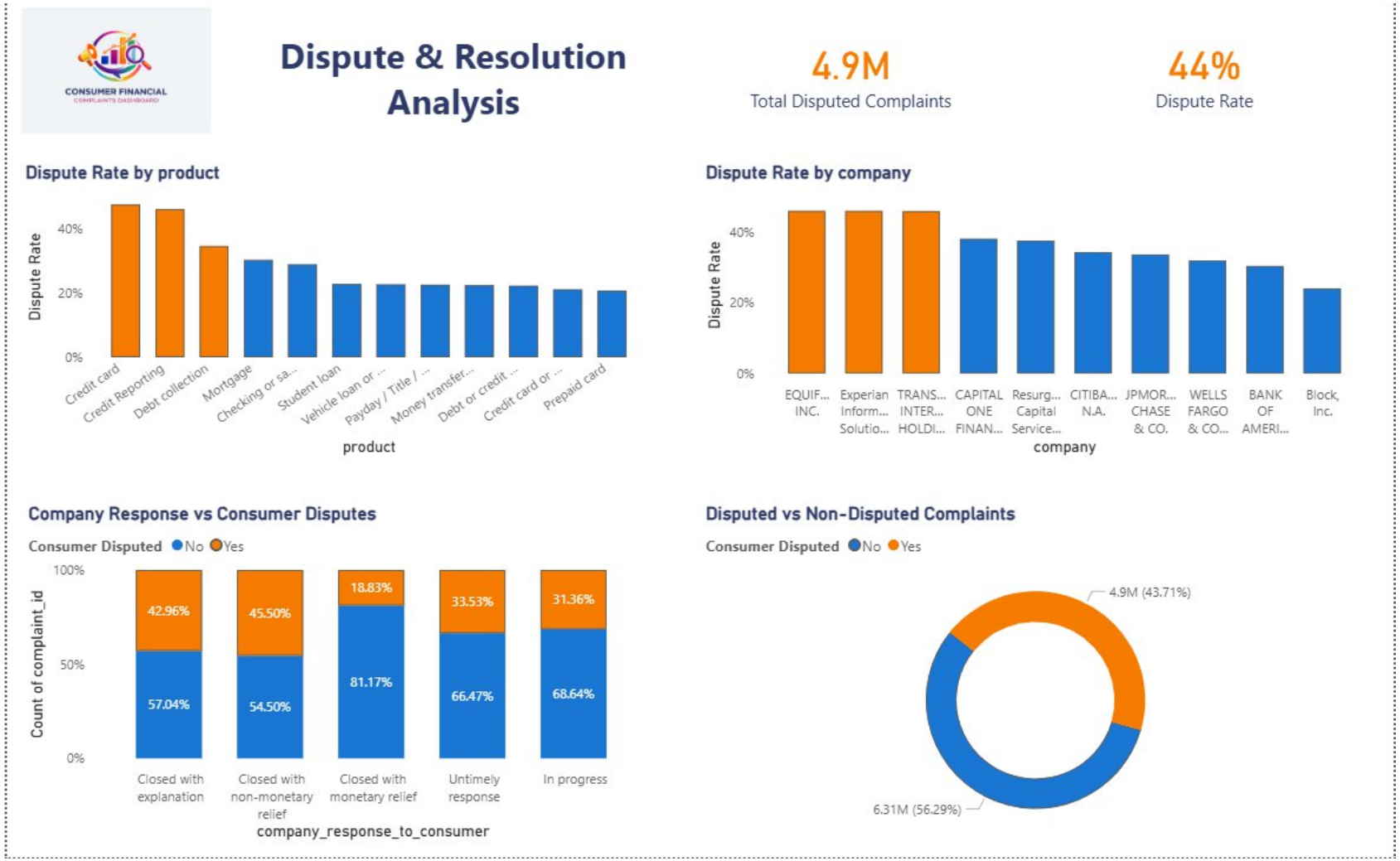
Key insights include:

- Dispute rates across financial products and companies

- Overall proportion of complaints that remain disputed

- Impact of company response types on dispute outcomes

The analysis shows that complaints resolved with **monetary relief have significantly lower dispute rates**, indicating more effective complaint resolution.



Key Insights:

Credit reporting and credit cards generate the highest complaint volumes, indicating major consumer concerns in these product categories.

The most frequently reported issue involves **incorrect information on credit reports**, suggesting data accuracy challenges.

Certain companies receive significantly higher complaint volumes, highlighting areas requiring regulatory attention.

Approximately **44% of complaints remain disputed**, indicating that many company responses do not fully resolve consumer concerns.

Complaints resolved with **monetary relief show the lowest dispute rates**, suggesting financial compensation is the most effective resolution strategy.

Recommendations:

Based on the analysis, the following actions are recommended:

1. Improve complaint resolution processes

2. Increase transparency in credit reporting systems

3. Strengthen oversight of debt collection practices

4. Encourage companies to prioritize **effective resolutions over explanatory responses**

Using complaint analytics regularly can help identify **emerging risks and improve consumer protection**.

Future Scope

1. Integrate real-time complaint data updates for continuous monitoring.

2. Apply predictive analytics to forecast complaint trends in financial products.

3. Expand the dashboard with geographical analysis and regulatory risk indicators.

4. Develop automated reporting for regulators and financial institutions.

Conclusion

This project analyzed the CFPB Consumer Financial Complaints dataset to understand patterns in consumer issues related to financial products and company responses. The Power BI dashboards effectively transformed a large complaint dataset into clear visual insights, helping identify major complaint drivers, common issue categories, and dispute trends across companies. The proposed solution proved effective in presenting complex complaint data in a structured and interactive format, allowing stakeholders to easily explore complaint patterns and identify areas of concern.

During the implementation, challenges were encountered while preparing the dataset, such as handling missing values, removing irrelevant fields, and structuring the data for efficient analysis. Ensuring that the dataset was properly cleaned and modeled for Power BI was essential for accurate visualization and meaningful insights.

Overall, the project demonstrates how data analytics and visualization tools can help interpret large complaint datasets and highlight key consumer issues in the financial sector. These insights can support better understanding of complaint trends and contribute to improving complaint resolution processes and consumer protection efforts.

REFERENCES

Dataset Source
Consumer Financial Protection Bureau (CFPB)
https://www.consumerfinance.gov/data-research/consumer-complaints/

Tools Used
PostgreSQL – Data cleaning and preparation
Power BI Desktop – Visualization and dashboard development
Power BI Service – Dashboard publishing and sharing

Project Repository
GitHub:
https://github.com/VarunJoshi-Max/Consumer-Financial-Complaints-Analysis

Power BI Dashboard
Power BI Service Report Link:
https://app.powerbi.com/links/f-TQpYn-tw?ctid=5cf28a99-8d4b-415c-a59e-cd6a57d65be6&pbi_source=linkShare

Power BI Project File (.pbix)
Google Drive Download:
https://drive.google.com/file/d/1TZenWRsfZURYYap5_ss7kJ0c5toIAYMo/view?usp=drive_link

Note: The Power BI (.pbix) file exceeds GitHub's upload size limit, therefore it has been shared via Google Drive.

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Thank You